

Awais Farhan Alam

Software Engineer • Full Stack Developer • DevOps & Cloud Engineer

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PROFESSIONAL SUMMARY

Motivated and results-driven Software Engineering student at NUTECH Islamabad (CGPA 3.4) with a strong foundation in full-stack web development, DevOps engineering, and cloud infrastructure. Proficient in building scalable web applications from concept to deployment using modern frameworks and toolchains. Skilled in containerization, CI/CD pipelines, Linux administration, and database management (SQL & NoSQL). Combines a disciplined academic background — including Matric (1059/1100) from APS FF Centre and ICS (945/1100) from APS Boys Pasban — with hands-on project delivery across multiple domains. Passionate about contributing to efficient, well-architected software systems while continuously growing in emerging technologies.

EDUCATION

BS Software Engineering — CGPA 3.4

2023 – Present

NUTECH — National University of Technology, Islamabad

Relevant coursework: Object-Oriented Programming, Data Structures & Algorithms, Database Systems, Web Engineering, Software Project Management, Software Quality Assurance, Operating Systems, Computer Networks.

ICS — Intermediate in Computer Science (945 / 1100)

2021 – 2023

APS Boys Pasban, Rawalpindi

Matric (1059 / 1100)

2019 – 2021

APS FF Centre, Abbottabad

WORK EXPERIENCE

Teaching & Academic Professional Work

2023 – Present

NUTECH — National University of Technology, Islamabad

Actively engaged in teaching assistance and collaborative academic research alongside full-time BS Software Engineering studies. Supported peers in technical subjects, assisted in lab sessions, and contributed to knowledge-sharing within the department.

- ▶ Mentored junior-year students in core programming subjects including C++, Data Structures, and Web Engineering.
- ▶ Assisted in evaluating lab exercises and guided students through debugging and problem-solving.
- ▶ Collaborated on group projects, taking technical leadership roles in system architecture and deployment decisions.
- ▶ Maintained consistent high academic performance (CGPA 3.4) while managing parallel project workloads.

PROJECTS

1. AI Model to Enhance Underwater Images

Tech Stack: Python, OpenCV, TensorFlow, NumPy, Matplotlib

Developed a deep learning pipeline that improves the visual quality of degraded underwater photographs using image-processing and neural network techniques. Underwater images suffer from colour distortion, low contrast, and blurring due to light scattering; this model corrects those artefacts automatically.

- ▶ Designed and trained CNN for image enhancement using paired underwater/clear dataset.
- ▶ Achieved measurable improvement in PSNR and SSIM metrics against baseline images.
- ▶ Built a Python CLI tool and a basic Streamlit-based web interface for demo purposes.

2. University Management System — Online

Tech Stack: *Java , JavaFX , MySQL*

Full-stack web application providing a centralised platform for university administration, faculty, and students. The system handles student enrolment, attendance, course management, and announcements under single roof.

- ▶ Designed a role-based access control (RBAC) system with three portals: Admin, Faculty, and Student.
- ▶ Desktop application built with Java/C++ and JavaFX/Windows Forms providing three role-based portals — Admin, Faculty, and Student — for managing enrolment, attendance, grades, scheduling, and announcements.
- ▶ MySQL relational database backend with normalized schemas for users, courses, departments, fee records, and grade history; Admin panel handles semester setup, user accounts, and department management.

3. Job Portal with Ciphers Encryption / Decryption

Tech Stack: *Node.js, Express.js, React.js, MongoDB, Crypto.js, AES / RSA*

A secure job-portal web application where candidate profiles, CVs, and communication between employers and applicants are protected using symmetric and asymmetric cryptographic algorithms. Designed to address real-world data-privacy concerns in online recruitment.

- ▶ Used RSA public/private key pairs for secure message exchange between employers and applicants.
- ▶ Built job listing, application tracking, and shortlisting modules with full CRUD operations.
- ▶ Admin panel for managing employer accounts, flagging suspicious listings, and auditing encrypted records.

4. Web-Based Chat Application (Full-Featured)

Tech Stack: *React.js, Node.js, Socket.IO, Express.js, MongoDB, Tailwind CSS*

A real-time, full-featured web chat application supporting one-on-one messaging, group chat rooms, media sharing, and user presence indicators. Built to match modern messaging UX standards.

- ▶ Features include: user registration/login, friend requests, group rooms, read receipts, and online/offline status.
- ▶ Media sharing (images and files) is handled via Multer with cloud storage integration.
- ▶ Messages stored persistently in MongoDB with pagination support for chat history.
- ▶ End-to-end message encryption layer using AES applied per conversation thread.

5. Cloud Infrastructure Deployment

Tech Stack: *AWS EC2 / S3 / RDS, Terraform, Docker, Nginx, Linux (Ubuntu), Bash*

Architected and provisioned a scalable cloud infrastructure on AWS to host multiple web applications with high availability, automated scaling, and secure networking.

- ▶ Configured Nginx as a reverse proxy with SSL/TLS termination using Let's Encrypt certificates.
- ▶ Deployed Dockerized services with Docker; implemented container restart policies and health checks.
- ▶ Stored static assets and backups on AWS S3 with lifecycle policies for cost optimisation.
- ▶ Set up AWS RDS (PostgreSQL) with automated snapshots and read-replica for database high availability.
- ▶ Wrote Bash scripts for server setup automation, log rotation, and system monitoring alerts.

6. Deployed Web Applications (Multi-Project Deployment)

Tech Stack: *Vercel, Netlify, VPS (DigitalOcean / AWS), Docker, PM2, Nginx, GitHub Actions*

Managed end-to-end deployment pipelines for multiple web applications across different hosting platforms, ensuring zero-downtime releases and environment parity.

- ▶ Set up CI/CD pipelines using GitHub Actions for automated build, test, and deploy workflows.
- ▶ Deployed Node.js/Express back-ends on VPS using PM2 with environment variable management.
- ▶ Hosted React/Next.js front-ends on Vercel and AWS with custom domain configuration, HTTPS enforcement.
- ▶ Configured Nginx virtual hosts for routing multiple apps on a single VPS using sub-domains.

7. Inventory Management System

Tech Stack: *React.js, Node.js, Express.js, MySQL, REST API*

A comprehensive web-based inventory and stock management system designed for small-to-medium businesses. Provides real-time tracking of products, suppliers, purchases, and sales with reporting dashboards.

- ▶ Built a product catalogue with category management, SKU tracking, and multi-warehouse support.
- ▶ Purchase Order and Sales Order workflow with status tracking (pending, approved, dispatched, received).
- ▶ Stock-level alerts and auto-reorder notifications when inventory falls below a configurable threshold.
- ▶ Interactive dashboard with Chart.js visualizations for stock turnover, top-selling, and monthly purchase trends.

- Role-based access: Admin, Manager, and Staff with auditable action logs.

8. School Management System

Tech Stack: *Next.js, Node.js, PostgreSQL, Prisma ORM, Tailwind CSS, JWT*

A full-stack school administration platform covering student enrolment, teacher management, class scheduling, attendance, examinations, and fee management for K-12 educational institutions.

- Multi-role system: Super Admin, Principal, Teacher, Student, and Parent portals with tailored dashboards.
- Automated timetable generation based on teacher availability, classroom capacity, and subject constraints.
- Fee management: challan generation, online payment integration, due-date reminders, and defaulter reports.
- Prisma ORM with PostgreSQL ensured type-safe database queries and easy schema migrations.

9. E-Commerce Site Deployment

Tech Stack: *Next.js, Stripe API, MongoDB, Redux Toolkit, Cloudinary, Docker, Vercel*

End-to-end design, development, and deployment of a fully functional e-commerce platform with product management, secure checkout, order tracking, and an admin back-office.

- Product catalogue with categories, filters, search, pagination, and image galleries (Cloudinary CDN).
- Shopping cart and wishlist with persistent state management using Redux Toolkit + localStorage.
- Integrated Stripe Payment Gateway for credit/debit card processing with webhook-based order confirmation.
- Admin dashboard for product CRUD, order fulfilment, inventory monitoring, and sales analytics.

SKILLS & TOOLS

Frontend	HTML5, CSS3, JavaScript (ES6+), React.js, Next.js, Vite, Tailwind CSS, Redux Toolkit
Backend	Node.js, Express.js, REST API Design, Socket.IO, JWT Authentication
Languages	JavaScript, Python, C++
DevOps	Docker, Docker Compose, Nginx, PM2, CI/CD (GitHub Actions), Linux (Ubuntu), Bash Scripting
Cloud	AWS (EC2, S3, RDS, VPC), Vercel, Netlify
Databases	MongoDB, PostgreSQL, MySQL, Prisma ORM, Redis (basic)
Version Ctrl	Git, GitHub, Branching Strategies, Pull Requests, Code Review (Coderabbit)
Security	AES-256 Encryption, RSA Key Pairs, JWT, HTTPS/TLS, Environment Variable Management
Tools	VS Code, Postman, Figma (basic), Webpack, npm/yarn
Soft Skills	Technical Leadership, Team Collaboration, Communication, Problem Solving, Time Management
Languages	English (Fluent), Urdu (Native)

3.4 CGPA	9+ Projects	2+ Years Experience	100% Commitment
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